1 A student is investigating simple harmonic motion using an electric vibrator. A plate is attached to the top of the electric vibrator. A small mass is placed on the metal plate as shown in Fig. 1.1.

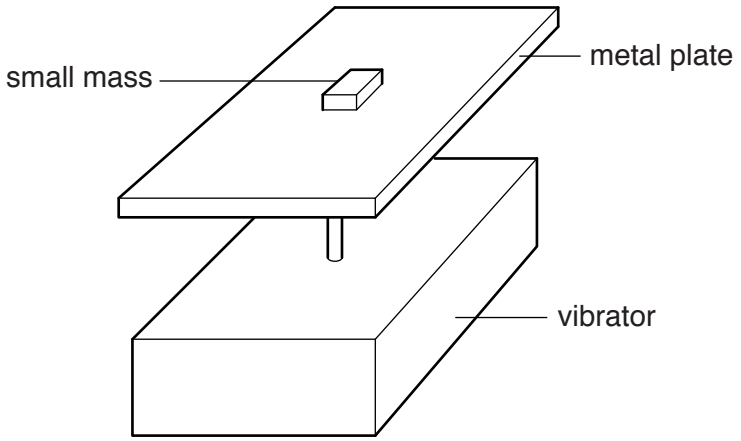

Fig. 1.1

An alternating potential difference (p.d.) is applied to the vibrator. For a given peak p.d. V, there is a maximum frequency f at which the small mass remains in contact with the plate. The contact between the small mass and plate is lost when the frequency is greater than f.

It is suggested that the relationship between f and V is

$$k = \pi^2 f^2 V$$

where k is a constant.

Design a laboratory experiment to test the relationship between f and V. Explain how your results could be used to determine a value for k. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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